Scott complexity of countable abelian p-groups: classification in the reduced case and the realization spectrum

21 October 2026
Working manuscript

Abstract

We determine the exact Scott complexity of every countable reduced abelian p-group. Write its length as $\omega\gamma + n$, $n < \omega$. At a limit length the complexity is $\Pi_{2\gamma+1}$; otherwise the bounded terminal subgroup at $\omega\gamma$ determines one of four exact complexities. The distinction depends on its infinite cyclic multiplicities and on shorter finite summands, rather than on how many infinite Ulm invariants occur anywhere in the group. We construct reduced examples at every positive countable successor difference-of-Σ level. We also exclude exact pure Σ complexity for every countable abelian p-group, including those with divisible summands. This answers Questions 5.1 and 5.2 of Alvir–Csima–MacLean in their stated versions. Their reduced-group back-and-forth theorem, Ulm classification, Zippin existence, and general Scott-complexity restrictions are credited inputs. The classification for individual nonreduced groups remains open here.

1 The problem, its source, and a first example

A Scott sentence describes one countable structure up to isomorphism among countable models. Its complexity measures how hard that description must be in countable infinitary logic. Algebraic classification alone does not settle the optimal logical complexity: it supplies descriptions, but one must still show that simpler descriptions cannot work.

Alvir, Csima and MacLean [1] give upper bounds and extensive back-and-forth analysis for reduced abelian p-groups. Their Question 5.1 asks about optimality when at least two Ulm invariants are infinite. Their Question 5.2 asks which pure Σ and difference complexities occur. Gonzalez, Rossegger and Turetsky [4] later ask for explicit abelian p-groups displaying a gap between two Scott rank conventions and for the non-Π spectrum. Their abstract existence argument is prior work, rather than a new conclusion here.

The simplest useful example is

$$C_p \oplus C_{p^2}^{(\omega)}.$$

It has exact d-Σ_2 complexity. The finite C_p summand can be described existentially, but it can disappear in a suitable increasing union of copies of the group, obstructing a Π_2 Scott sentence. If the C_p summand is absent, the complexity drops to Π_2. If both multiplicities are infinite, it rises to Π_3. The same three-way distinction reappears at any terminal ω-block of the transfinite p-filtration, with a separate finite-terminal case. The classification below makes that assertion exact.

The proof has three main steps. First we give the bounded descriptions. Next we construct quotients which change a terminal invariant while preserving specified back-and-forth

comparisons. The same quotient must serve both comparisons in the highest-complexity case; separate competitors would not exclude a difference sentence. Finally we build explicit transfinite examples and handle the divisible part separately.

2 Conventions and the classification

Fix a prime p. Groups are countable abelian p-groups in the original language $\{+,-,0\}$. No filtration predicates are added to that language. Write C_{p^j} for the cyclic group of order p^j and C_{p^∞} for the Prüfer group. The standard decomposition is $G = R \oplus D$, with R reduced and D divisible. Define

$$G_0 = G, \qquad G_{\xi+1} = pG_\xi, \qquad G_\lambda = \bigcap_{\xi<\lambda} G_\xi.$$

$$P_\xi(G) = G_\xi[p], \qquad u_G(\xi) = \dim_{\mathbb{F}_p} P_\xi(G)/P_{\xi+1}(G).$$

The reduced length λ_R is the least index with $R_{\lambda_R} = 0$. For $x \neq 0$, its height is ξ when $x \in G_\xi \setminus G_{\xi+1}$; elements in the divisible part and zero have height ∞. All ordinal products are ordinary ordinal products: $2\gamma = 2 \cdot \gamma$ and $\omega\gamma = \omega \cdot \gamma$.

We use ordinary, not necessarily computable, infinitary Σ_α and Π_α formulas. At level zero they are quantifier-free; a Σ_α formula is a countable disjunction of existential formulas with Π_β matrices, $\beta < \alpha$, and Π_α is the dual class. A d-Σ_α sentence is a conjunction of a Σ_α and a Π_α sentence. *Exact* complexity refers to the usual Scott-complexity hierarchy; it does not deny that a structure has padded descriptions at higher levels. We use the general restrictions and Miller collapse from [2, Theorems 1.6, 2.5 and 2.6].

For finite tuples, $(A, \bar{a}) \leq_0 (B, \bar{b})$ means atomic agreement. For $\alpha > 0$, every $\beta < \alpha$ and finite right extension $\bar{d}$ must have a left response $\bar{c}$ with $(B, \bar{b}\bar{d}) \leq_\beta (A, \bar{a}\bar{c})$. Thus Π_α truth transfers from the left to the right, and Σ_α truth transfers from the right to the left. This direction will be used throughout.

Theorem 2.1 (Exact reduced-group classification)**.** *Let $G \neq 0$ be reduced, and write its length uniquely as $\lambda = \omega\gamma + n$, where $n < \omega$. If $n = 0$, then $\gamma > 0$ and G has exact complexity $\Pi_{2\gamma+1}$. If $n > 0$, put $H = G_{\omega\gamma}$ and write*

$$H \cong \bigoplus_{j=1}^{n} C_{p^j}^{(m_j)}, \qquad m_j \in \mathbb{N} \cup \{\omega\}, \quad m_n > 0.$$

The exact complexity is

Condition on H	Exact complexity
H is finite	d-$\Sigma_{2\gamma+1}$
Exactly one $m_k = \omega$, and $m_j = 0$ for every $j < k$	$\Pi_{2\gamma+2}$
Exactly one $m_k = \omega$, and $m_j > 0$ for some $j < k$	d-$\Sigma_{2\gamma+2}$
At least two $m_j = \omega$	$\Pi_{2\gamma+3}$

The trivial group has exact complexity Π_1.

The terminal multiplicities are those of the bounded group $G_{\omega\gamma}$, not the complete list of Ulm invariants of G. For example, finite summands above its unique infinite multiplicity are allowed in both middle rows. Their presence does not change which row applies.

3 The bounded base case

Theorem 3.1. *Let G be a countably infinite Abelian p-group of bounded exponent, and write*

$$G \cong \bigoplus_{j=1}^{n} C_{p^j}^{(m_j)}, \qquad m_j \in \mathbb{N} \cup \{\omega\}.$$

Its Scott complexity is determined as follows.

Condition	*Exact complexity*
At least two m_j equal ω	Π_3
Exactly one $m_k = \omega$, and some $m_j > 0$ with $j < k$	$d\text{-}\Sigma_2$
Exactly one $m_k = \omega$, and all $m_j = 0$ with $j < k$	Π_2

Thus no bounded-exponent countable Abelian p-group has an exact pure Σ Scott complexity. The case of at least two infinite multiplicities is [1, Theorem 4.2]; the proof below supplies the remaining cases, including arbitrary finite summands above the unique infinite multiplicity.

Proof. The decomposition is the standard bounded-exponent case of the Ulm classification. Suppose that the only infinite multiplicity is at k, and write

$$G = F_- \oplus C_{p^k}^{(\omega)} \oplus F_+,$$

where F_- has cyclic summands of orders below p^k, and F_+ has cyclic summands of orders above p^k. Both F_- and F_+ are finite. Let $K = p^k F_+$, a fixed finite group.

The condition $p^k G \cong K$ has a d-Σ_1 description. If K has M elements, an existential tuple of elements $p^k y_1, \ldots, p^k y_M$ can carry its complete finite group table, including all required inequalities. This is Σ_1. The bound $|p^k G| \leq M$ is Π_1, expressed by

$$\forall y_0 \cdots y_M \bigvee_{0 \leq i < j \leq M} p^k y_i = p^k y_j.$$

Together these conditions fix all cyclic multiplicities above k. The condition that $p^{k-1}G$ is infinite is Π_2:

$$\bigwedge_{m \geq 1} \exists y_1 \cdots y_m \bigwedge_{i<j} p^{k-1} y_i \neq p^{k-1} y_j.$$

Together with finiteness of $p^k G$, it forces the multiplicity at k to be ω.

For each $j < k$, the specified finite multiplicity m_j is a finite Ulm invariant. The usual lower dimension condition is Σ_2, and the corresponding upper condition is Π_2: choose elements in $(p^{j-1}G)[p]$ and require their nonzero linear combinations to lie outside $p^j G$. A finite conjunction of these exact dimension conditions is d-Σ_2. Add the Abelian group axioms and the bound $p^n G = 0$. These conditions determine the entire cyclic decomposition, and hence give a d-Σ_2 Scott sentence.

If $F_- = 0$, replace all the lower dimension conditions by

$$\forall x \left(px = 0 \;\longrightarrow\; \exists y \, p^{k-1} y = x \right).$$

This Π_2 sentence says that every socle element has height at least $k-1$, and so there are no cyclic summands shorter than p^k. The remaining conditions above are all Π_2 or of lower complexity. Thus G has a Π_2 Scott sentence in this case. It is exact: an infinite group cannot

have a universal Scott sentence because such a sentence passes to the trivial subgroup; a d-Σ_1 sentence would pass to the finite subgroup generated by its finite existential witnesses.

If $F_- \neq 0$, no Σ_2 Scott sentence is possible. More generally, no infinite locally finite group has a Σ_2 Scott sentence. One true existential disjunct has finite witnesses and a Π_1 matrix. The witnesses generate a finite subgroup, and the matrix passes to that subgroup. The purported Scott sentence would therefore have a nonisomorphic finite countable model.

To exclude Π_2 in the case $F_- \neq 0$, choose cyclic generators $u_1, \dots, u_t$ for F_- with orders $p^{\ell_1}, \dots, p^{\ell_t}$, where $1 \leq \ell_j < k$, and generators $e_0, e_1, \dots$ for the C_{p^k} summands. Define an injective endomorphism by

$$f(u_j) = p^{k-\ell_j} e_{j-1}, \qquad f(e_i) = e_{i+t}, \qquad f \restriction F_+ = \mathrm{id}\,.$$

Orders are preserved and the specified summand images have disjoint supports. In its direct limit, the finite F_+ factor stays fixed. The remaining factor has exponent p^k, an infinite p^{k-1}-multiple subgroup, and every socle element has a p^{k-1}-root at a later stage: one application of f moves the socle of each short summand into $p^{k-1}\langle e_{j-1}\rangle$. The cyclic decomposition therefore makes the limit

$$D \cong C_{p^k}^{(\omega)} \oplus F_+,$$

which is nonisomorphic to G because its lower multiplicities are all zero. A Π_2 sentence is preserved in an increasing union of substructures: every finite universal tuple occurs at a stage, whose finite existential witnesses and quantifier-free diagram remain in the union. Thus no Π_2 Scott sentence for G exists. The d-Σ_2 upper bound is consequently exact.

The only remaining infinite case has at least two infinite multiplicities, whose exact Π_3 complexity is the cited theorem of Alvir–Csima–MacLean. Finite nontrivial groups have exact d-Σ_1, and the trivial group has a Π_1 Scott sentence, so finite groups add no pure Σ exception. □

4 Changing tails while controlling comparisons

We use Ulm classification and the comparison theorem [1, Theorem 4.4] for reduced groups. Its odd and even cases include alternatives expressed by relative Σ_1 and Σ_2 truth in the terminal subgroup. We check those alternatives explicitly; they are not discarded when a terminal socle is finite.

A second input is Zippin's existence theorem, in the form stated by Calderoni and Thomas [3, Theorem 4.2]: a countable sequence of nonzero countable direct sums of cyclic p-groups is the Ulm-factor sequence of a reduced group exactly when every nonfinal factor has unbounded exponent. Here the factors are $U_\xi(G) = G_{\omega\xi}/G_{\omega(\xi+1)}$. We use that published theorem rather than a proposed infinite reverse iteration of one-block constructions.

Lemma 4.1 (Transfinite prescribed-tail existence)**.** *Let $\gamma > 0$ be countable, put $\delta = \omega\gamma$, and let G be a countable reduced Abelian p-group of length at least δ. For every nonzero countable reduced p-group T there is a countable reduced p-group $H(T)$ such that*

$$u_{H(T)}(\xi) = u_G(\xi) \quad (\xi < \delta), \qquad H(T)_\delta \cong T.$$

Its length is $\delta + \lambda_T$.

Proof. For each $\xi < \gamma$, $G_{\omega\xi}$ has length at least ω. Its nonzero finite Ulm invariants have unbounded support: otherwise a finite-stage subgroup would be divisible, contradicting reducedness. Consequently every $U_\xi(G)$ with $\xi < \gamma$ is nonzero and has unbounded exponent, including the last one when γ is a successor.

Let $\tau(T)$ be the ordinal length of the nonzero Ulm-factor sequence of T. Concatenate the first γ Ulm factors of G with the entire Ulm factor sequence of T. This is a countable sequence indexed by $\gamma+\tau(T)$. Every nonfinal factor is unbounded: this has just been checked for the initial factors, and it holds for the nonfinal factors of the reduced group T. Zippin's theorem supplies a reduced group H with this sequence. The Ulm factors of H_δ are exactly those of T, so Ulm's classification gives $H_\delta \cong T$. The initial multiplicities give the displayed lower Ulm identities. Finally $H_{\delta+\eta} = (H_\delta)_\eta$ and no earlier initial factor vanishes, giving length $\delta + \lambda_T$. □

Lemma 4.2 (Quotient filtration below a limit)**.** *Let $\delta = \omega\gamma$ and $T \leq G_\delta$. For the quotient map $q : G \to Q = G/T$, one has*

$$Q_\alpha = q(G_\alpha) \quad (\alpha \leq \delta).$$

For every $\beta < \delta$, q induces an isomorphism of the β-th Ulm quotients, so $u_Q(\beta) = u_G(\beta)$.

Proof. The successor step follows from $pq(G_\alpha) = q(pG_\alpha)$. At a limit $\eta \leq \delta$, the subgroup T lies in every G_α, $\alpha < \eta$, so $q^{-1}(q(G_\alpha)) = G_\alpha$. Intersecting preimages proves the limit step, including $\eta = \delta$.

For the Ulm quotient, $\beta + 2 < \delta$ whenever $\beta < \delta$; the case $\delta = 0$ has no such β. Let $y \in P_\beta(Q)$ and choose $x \in G_\beta$ with $q(x) = y$. Then $px \in T \subseteq G_{\beta+2}$, so there is $z \in G_{\beta+1}$ with $pz = px$. Thus $x - z \in P_\beta(G)$ and $q(x - z)$ differs from y by the element $q(z) \in P_{\beta+1}(Q)$. This gives surjectivity on Ulm quotients. If $x \in P_\beta(G)$ and $q(x) \in P_{\beta+1}(Q)$, then $x \in G_{\beta+1} + T = G_{\beta+1}$, proving injectivity. □

Lemma 4.3 (Finite-product preservation)**.** *If $(A_i, \bar{a}_i) \leq_\alpha (B_i, \bar{b}_i)$ for $i = 1, 2$, then*

$$(A_1 \oplus A_2, (\bar{a}_1, \bar{a}_2)) \leq_\alpha (B_1 \oplus B_2, (\bar{b}_1, \bar{b}_2)).$$

The paired notation denotes componentwise pairing of corresponding entries into tuples of product elements.

Proof. Induct on α. Equations between group terms in the product hold exactly when they hold in both coordinates, so atomic agreement is preserved. Given a right-hand finite extension and $\beta < \alpha$, split it into its coordinate tuples. Apply the two given comparisons to obtain coordinate responses. The induction hypothesis at β combines their reversed comparisons into the response in the product. This proves the assertion at successors and limits alike. □

4.1 The obstruction at odd-form levels

The explicit groups B_ω, W_ω used next are defined and proved to have the indicated filtrations in Section 6.

Theorem 4.4. *Let $\gamma > 0$ be countable. If $G = R \oplus D$, with R reduced and D divisible, and $\lambda_R \geq \omega\gamma$, then G has no $\Sigma_{2\gamma+1}$ Scott sentence. No countable Abelian p-group has exact pure $\Sigma_{2\gamma+1}$ complexity.*

Proof. Put $\delta = \omega\gamma$. Take the two reduced tails

$$T_0 = \bigoplus_{n<\omega} C^{(\omega)}_{p^{n+1}}, \qquad T_1 = B_\omega \oplus W_\omega,$$

where B_ω, W_ω are the presentations in Section 6. Their lengths are respectively ω and $\omega+1$, and their socles at every finite height are infinite. Lemma 4.1 gives $H_i = H(T_i)$ with the same Ulm invariants as R below δ and with distinct lengths $\delta+\omega$, $\delta+\omega+1$.

Apply ACM, Theorem 4.4(1) [1], with left group H_i, right group R, index γ, and empty tuples. Condition (a) is the equality of lower invariants. Condition (b) holds because

$$\lambda_{H_i} \geq \delta + \omega \geq \min\{\lambda_R, \delta+\omega\}.$$

For condition (c), every $P_\xi(H_i)$ with $\xi < \delta+\omega$ is infinite. Below δ it contains the infinite socle of the tail; above δ use its finite-height socles. Conditions (d),(e) are vacuous for the empty tuple. Therefore

$$H_i \leq_{2\gamma+1} R.$$

Lemma 4.3 works at every ordinal, so $H_i \oplus D \leq_{2\gamma+1} R \oplus D$. Every $\Sigma_{2\gamma+1}$ sentence true of G is true in both $H_i \oplus D$. Their reduced parts have different lengths, so at least one is nonisomorphic to G. This proves the first assertion.

Suppose now that $\lambda_R < \omega\gamma$. If $\gamma = \beta+1$, then $G_{\omega\beta}$ stabilizes after finitely many p-steps. Such a terminal group has a Π_3 Scott sentence: finite-stage stabilization, its finite-height Ulm invariants, and the rank of its divisible part can all be specified at that level. Relativization to $G_{\omega\beta}$, a $\Pi_{2\beta}$ domain, gives a $\Pi_{2\beta+3} = \Pi_{2\gamma+1}$ sentence. The lower invariant conditions have complexity at most $\Pi_{2\beta+1}$, so they fit. The ambient Abelian p-group axioms are included.

If γ is a limit, choose $\beta < \gamma$ and $n < \omega$ with $\lambda_R = \omega\beta + n$. The same finite-stabilization argument gives a $\Pi_{2\beta+3}$ Scott sentence; now $2\beta+3 < 2\gamma$. Thus in both cases there is a $\Pi_{2\gamma+1}$ Scott sentence. If a $\Sigma_{2\gamma+1}$ Scott sentence also exists, Miller's theorem in AGHT, Theorem 2.5, gives a d-Σ_η Scott sentence for some $\eta < 2\gamma+1$ (in particular one at 2γ). Hence pure $\Sigma_{2\gamma+1}$ cannot be exact. □

4.2 Deleting a direct summand of a bounded tail

Lemma 4.5 (Finite support absorption)**.** *Suppose $H = K \oplus T$ is bounded, $p^k T = 0$, and a cyclic decomposition of K contains infinitely many C_{p^k} summands. Then $K \preceq_{\Sigma_1} H$, for ordinary infinitary formulas with parameters from K. Consequently $H \leq_2 K$.*

Proof. Existential truth passes from K to H by the subgroup inclusion. Conversely, fix finite parameters in K and finite witnesses in H. Their K coordinates and the parameters lie in a finite direct summand K_0 of K. Their T coordinates generate a finite group T_0. Write T_0 as a finite sum of cyclic groups of orders p^{ℓ_i}, where $\ell_i \leq k$. Choose distinct C_{p^k} coordinates of K outside the support of K_0, with generators e_i, and send the i-th cyclic generator of T_0 to $p^{k-\ell_i}e_i$. Together with the identity on K_0, this embeds $K_0 \oplus T_0$ into K and fixes the parameters. It preserves equations and inequations, so the witnesses can be moved into K. This applies to any true existential disjunct of an infinitary Σ_1 formula.

Hence $(K, \bar{b}) \leq_1 (H, \bar{b})$ for every finite tuple $\bar{b}$ in K. To verify $H \leq_2 K$, answer each such tuple by itself in H. The level-1 requirement is the displayed comparison, and the level-0 requirement follows as well. □

The exponent bound is essential: an order-p^{k+1} cyclic group cannot be embedded into a fresh order-p^k coordinate. Every finite higher summand is retained in the applications below.

Lemma 4.6 (Deleting a terminal direct summand)**.** *Put $\delta = \omega\gamma$, where $\gamma \geq 0$, and suppose G is reduced with nonzero bounded $H = G_\delta$. Write $H = K \oplus T$, assume K has the same exponent as H, and put $Q = G/T$, with quotient map q. For $A = \langle \bar{a} \rangle$ finite, suppose $A \cap H \subseteq K$. Then*

$$u_Q(\xi) = u_G(\xi) \quad (\xi < \delta), \qquad Q_\delta \cong K, \qquad \lambda_Q = \lambda_G,$$

and

$$(G, \bar{a}) \leq_{2\gamma+1} (Q, q\bar{a}). \tag{1}$$

If also $H \leq_2 K$, then

$$G \leq_{2\gamma+2} Q. \tag{2}$$

Proof. Lemma 4.2 applies to every $T \leq G_\delta$: $Q_\xi = q(G_\xi)$ through δ, and all lower Ulm quotients are unchanged. Thus $Q_\delta = H/T \cong K$. Equal exponents give equal lengths and ensure Q is reduced.

The pointed argument extends the finite retained-summand construction to arbitrary K. The map q is injective on $A + K$: if $a + x \in T$, where $a \in A$ and $x \in K$, then $a \in A \cap H \subseteq K$, so $a + x \in K \cap T = 0$. An element of A outside H retains its height by taking inverse images of the quotient filtration through δ. For an element of $A \cap H$, height inside H equals height in its direct summand K; adding δ shows the full height is also unchanged. Zero retains height infinity.

Apply ACM Theorem 4.4(1) to left G, right Q, and the inverse partial map induced by q. Lower invariants, lengths, generated subgroups, and heights meet clauses (a), (b), (d), and (e). For (c), lift witnesses of any relative Σ_1 formula from $Q_\delta = qK$ to K. Injectivity on $A + K$ preserves every positive and negative atomic formula, including equations involving parameters outside H. This is the required transfer from $(Q, q\bar{a})$ to $(G, \bar{a})$, proving (1).

For (2), apply ACM Theorem 4.4(2) with empty tuples. Lower invariants agree. In the next block, the invariants of Q are bounded by those of G, since K is a direct summand of H. Lengths agree below $\delta + \omega$. In clause (c), use $H \leq_2 K$: every Σ_2 sentence true in $K \cong Q_\delta$ holds in H. For empty tuples this is precisely the required relative clause. The subgroup and height clauses are vacuous. □

This is a sufficient algebraic criterion for the indicated pairs. It is not an elimination of ACM's exceptional clauses for all pairs and all distinguished tuples.

5 Proof of the reduced classification

Proof of Theorem 2.1. Put $\delta = \omega\gamma$. By [1, Proposition 2.2], membership in G_δ is $\Pi_{2\gamma}$-definable. Specifying the Ulm invariants below δ costs at most $\Pi_{2\gamma+1}$: at $\omega\eta + r < \delta$, a dimension-at-least-m condition is $\Sigma_{2\eta+2}$, and an infinite dimension costs $\Pi_{2\eta+3}$, where $\eta < \gamma$.

For finite nonzero H and $\gamma > 0$, require a complete group table on $|H|$ distinct points of G_δ by a $\Sigma_{2\gamma+1}$ formula, and bounds the size of G_δ by a $\Pi_{2\gamma+1}$ formula. Along with lower invariants and the ambient Abelian p-group axioms, this gives d-$\Sigma_{2\gamma+1}$.

With exactly one infinite multiplicity, write

$$H = F_- \oplus C_{p^k}^{(\omega)} \oplus F_+,$$

where F_- and F_+ are finite sums of shorter and longer cyclic groups. Theorem 3.1 gives a Π_2 Scott sentence when $F_- = 0$ and a d-Σ_2 Scott sentence otherwise. Relativization to G_δ yields $\Pi_{2\gamma+2}$ and d-$\Sigma_{2\gamma+2}$ respectively. A relative existential block followed by a relative Π_1 matrix is $\Sigma_{2\gamma+2}$; the dual universal form is $\Pi_{2\gamma+2}$. The lower invariant conditions fit either bound.

With at least two infinite terminal multiplicities, ACM Theorem 3.1(ii) gives $\Pi_{2\gamma+3}$. With length exactly δ, Theorem 3.1(iii) gives $\Pi_{2\gamma+1}$. Each description forces the correct bounded tail, hence reducedness, and the complete Ulm sequence. Ulm classification gives countable categoricity. A nontrivial finite group ($\gamma = 0$ and finite H) instead uses its finite table and cardinality bound directly, giving d-Σ_1 without adjoining higher-complexity ambient axioms.

Lower bounds for infinite terminal subgroups.

The finite-K instance of Lemma 4.6 gives no $\Sigma_{2\gamma+2}$ Scott sentence for any reduced G with infinite bounded G_δ. It is the finite-K case of (1): choose a finite direct summand containing the parameters' intersection with H and a maximal-order coordinate. This also excludes every lower class below $\Pi_{2\gamma+2}$. Thus the $\Pi_{2\gamma+2}$ row with $F_- = 0$ is exact.

If $F_- \neq 0$, take $T = F_-$ and $K = C_{p^k}^{(\omega)} \oplus F_+$. The absorption lemma gives $H \leq_2 K$, and (2) gives

$$G \leq_{2\gamma+2} G/F_-.$$

A nonzero shorter terminal invariant has been removed, so the groups are nonisomorphic. Any $\Pi_{2\gamma+2}$ Scott sentence would transfer to this nonisomorphic quotient. The upper bound and the pointed quotient $\Sigma_{2\gamma+2}$ obstruction therefore give exact d-$\Sigma_{2\gamma+2}$.

Suppose now that $m_i = m_k = \omega$ for $i < k$, and fix any finite $\bar{a}$ in G. In a cyclic decomposition of H, retain the finitely many C_{p^i} coordinates supporting $A \cap H$, where $A = \langle \bar{a} \rangle$. Let T be the sum of all the other C_{p^i} coordinates, and let K retain all remaining coordinates, including every coordinate of order different from p^i. Then

$$H = K \oplus T, \qquad A \cap H \subseteq K, \qquad p^i T = 0, \qquad C_{p^k}^{(\omega)} \text{ is a direct summand of } K.$$

The exponent is unchanged. The two lemmas give, for $Q = G/T$,

$$(G, \bar{a}) \leq_{2\gamma+1} (Q, q\bar{a}), \qquad G \leq_{2\gamma+2} Q.$$

The invariant $u_Q(\delta + i - 1)$ is finite whereas $u_G(\delta + i - 1) = \omega$, so $Q \ncong G$.

If a d-$\Sigma_{2\gamma+2}$ Scott sentence were $S \wedge P$, choose $\bar{a}$ witnessing a true existential disjunct of its $\Sigma_{2\gamma+2}$ part S, with $\Pi_{2\gamma+1}$ matrix. The pointed comparison transfers this matrix to $(Q, q\bar{a})$; the unpointed comparison transfers its $\Pi_{2\gamma+2}$ part P. This contradicts categoricity. Hence no such difference Scott sentence exists. The $\Pi_{2\gamma+3}$ upper bound is exact. An additional $\Sigma_{2\gamma+3}$ Scott sentence would, by Miller collapse (AGHT Theorem 2.5 [2]), yield a lower difference description and is also impossible.

Finite tails and exact limit lengths.

Suppose $\gamma > 0$ and $0 < |G_\delta| < \omega$. Theorem 4.4 excludes a $\Sigma_{2\gamma+1}$ Scott sentence because the reduced length is at least δ. Put $Q = G/G_\delta$. The quotient lemma preserves the invariants below δ and gives length δ. To verify the latter equality, if $Q_\xi = 0$ for some $\xi < \delta$, then $G_\xi \subseteq G_\delta$. If $p^m G_\delta = 0$, this would give $G_{\xi+m} = 0$, where $\xi + m < \delta$, contradicting the length of G. ACM Theorem 4.4(1), with left G, right Q, and empty parameters, applies: the zero tail Q_δ embeds in G_δ, so its Σ_1 theory transfers in the required direction for clause (c). Therefore

$$G \leq_{2\gamma+1} Q.$$

Since $Q \ncong G$, this excludes $\Pi_{2\gamma+1}$ and proves the finite-terminal difference row. For nontrivial finite groups, existential truth passes to a proper larger group and universal truth passes to the trivial subgroup. Thus neither Σ_1 nor Π_1 is a Scott complexity, completing the $\gamma = 0$ finite case.

Finally suppose $\lambda_G = \delta = \omega\gamma$, where $\gamma > 0$. Lemma 4.1 supplies a reduced B with the same invariants below δ and $B_\delta \cong C_p$. Its length is $\delta + 1$, so it is nonisomorphic to G. We have

$$G \leq_{2\gamma} B \qquad \text{and} \qquad B \leq_{2\gamma} G.$$

If $\gamma = \eta + 1$, apply ACM Theorem 4.4(2) at index η in both directions. The invariants agree below $\delta = \omega\eta+\omega$, and both lengths are at least δ. Every socle below δ in each group is infinite: a finite socle in a reduced p-group makes the corresponding subgroup finite, forcing the full length below the next limit δ. The finite-socle fact is stated just after ACM Theorem 4.4 and also has a short direct proof. If a group with finite socle had unbounded exponent, some nonzero socle element would have roots of arbitrarily large finite depth. Each p-root fiber has at most the finite size of the socle. König's lemma would therefore give an infinite compatible root chain, generating a Prüfer subgroup and contradicting reducedness. Once the exponent is bounded by p^m, induction through the p-map, whose fibers are finite, proves that the group is finite. Thus the algebraic infinite-socle alternative in clause (c) applies on both sides. Empty parameters meet the remaining clauses. If γ is a nonzero limit, $2\gamma = \gamma$ and ACM Theorem 4.4(3) applies directly.

Consequently G has no d-$\Sigma_{2\gamma}$ Scott sentence. The $\Pi_{2\gamma+1}$ upper bound and Miller collapse prove exactness for successor as well as limit γ. The trivial group is described by $\forall x\,(x = 0)$ and has no quantifier-free Scott sentence. This proves every row of the classification.

□

6 Explicit successor realizations

For a countable ordinal ξ, let W_ξ be the Abelian group with a generator e_s for each nonempty finite strictly decreasing sequence

$$s = (\xi, \xi_1, \ldots, \xi_n), \qquad \xi > \xi_1 > \cdots > \xi_n,$$

including the one-term sequence (ξ). Its defining relations are

$$pe_{(\xi)} = 0, \qquad pe_{s^\frown\eta} = e_s \quad (\eta < \operatorname{last}(s)).$$

This is the usual decreasing-sequence presentation of a Walker group; the properties needed here follow directly from the presentation.

Lemma 6.1. *Every element of W_ξ has a unique finite expression*

$$\sum_s c_s e_s, \qquad 0 \leq c_s < p.$$

Moreover, for every ordinal α,

$$(W_\xi)_\alpha = \langle e_s : \operatorname{last}(s) \geq \alpha \rangle . \tag{3}$$

Consequently,

$$(W_\xi)_\xi = \langle e_{(\xi)} \rangle \cong C_p, \qquad (W_\xi)_{\xi+1} = 0.$$

For every ordinal δ,

$$(W_{\delta+1})_\delta \cong C_{p^2}.$$

Proof. A finite ancestor-closed set of sequences carries the finite presentation obtained by restricting the displayed relations to those sequences. Adding a new leaf s to such a finite set adjoins a root of its parent with relation $pe_s = e_{s^-}$. The old finite group embeds in this extension: in $(H \oplus \mathbb{Z})/\langle(-e_{s^-}, p)\rangle$, the relation subgroup meets $H \oplus \{0\}$ only at zero. The new quotient has exactly p cosets over the old group. The root itself starts with a cyclic group of order p. Induction over the finite tree therefore gives a unique expression with all coefficients between 0 and $p-1$. All finite ancestor-closed presentations embed in larger ones by the same argument, so the result holds in their directed union W_ξ. In particular, there are no further relations between digit-normal expressions. Each generator has finite p-power order, so the group is indeed a countable Abelian p-group.

Let S_α be the set of nodes with last entry at least α. It is ancestor-closed. An element belongs to $\langle S_\alpha\rangle$ exactly when its digit-normal support is contained in S_α. At a successor, the p-multiples of generators in S_α lie in $\langle S_{\alpha+1}\rangle$. Conversely, if $\mathrm{last}(s) \geq \alpha + 1$, the child $s \frown \alpha$ exists, belongs to S_α, and is a p-root of e_s. Thus

$$p\langle S_\alpha\rangle = \langle S_{\alpha+1}\rangle.$$

At a limit λ, uniqueness of finite digit support gives

$$\bigcap_{\alpha<\lambda} \langle S_\alpha\rangle = \langle S_\lambda\rangle.$$

Transfinite induction proves (3). At level ξ only the root remains. At level δ in $W_{\delta+1}$, exactly the root and its child $(\delta+1, \delta)$ remain; the latter has p-multiple equal to the former, giving a cyclic group of order p^2. □

Transfinite p-subgroups commute with direct sums:

$$\left(\bigoplus_i G_i\right)_\alpha = \bigoplus_i (G_i)_\alpha. \tag{4}$$

The successor step is immediate. At a limit, an element has finite coordinate support, and it belongs to each previous subgroup exactly when each coordinate does. This proves the identity by induction. The corresponding Ulm invariants add.

For $\delta = \omega\gamma$, define

$$B_\delta = \bigoplus_{\xi<\delta} W_\xi^{(\omega)}, \tag{5}$$

$$A^{\mathrm{odd}}_{\gamma,r} = B_\delta \oplus W_\delta^r, \tag{6}$$

$$A^{\mathrm{even}}_{\gamma,r} = B_\delta \oplus W_\delta^r \oplus W_{\delta+1}^{(\omega)}. \tag{7}$$

These are countable presentations for every countable γ. They need not be computable presentations when γ is noncomputable; no effectivity of arbitrary countable ordinal input is asserted.

Each copy of W_ξ contributes one dimension to its ξ-th Ulm invariant. The infinitely many copies in (5) therefore give

$$u_{B_\delta}(\xi) = \omega \quad (\xi < \delta), \qquad (B_\delta)_\delta = 0.$$

Lemma 6.1 now gives the complete invariant lists needed for the two families:

Group	$\xi < \delta$	$\xi = \delta$	$\xi = \delta + 1$	later ξ
$A^{\text{odd}}_{\gamma,r}$	ω	r	0	0
$A^{\text{even}}_{\gamma,r}$	ω	r	ω	0

Their lengths are $\delta + 1$ and $\delta + 2$, respectively, and

$$(A^{\text{odd}}_{\gamma,r})_\delta \cong C^r_p, \qquad (A^{\text{even}}_{\gamma,r})_\delta \cong C^r_p \oplus C^{(\omega)}_{p^2}. \tag{8}$$

Both are reduced because their transfinite p-filtrations reach zero. For different r, their δ-th Ulm invariants differ.

Theorem 6.2 (The difference spectrum)**.** *Every positive countable successor d-Σ_α complexity occurs among countable reduced abelian p-groups. Among infinite such groups the attained difference levels are exactly the successors $\alpha \geq 2$.*

Proof. For $\gamma > 0$ the two groups in (6) and (7) have terminal groups displayed in (8). Theorem 2.1 gives exact complexities d-$\Sigma_{2\gamma+1}$ and d-$\Sigma_{2\gamma+2}$ respectively. At level two take $C^r_p \oplus C^{(\omega)}_{p^2}$, $r \geq 1$; nontrivial finite groups give level one. Every successor ordinal is $2\gamma + 1$ or $2\gamma+2$: write it as a limit (or zero) plus a positive finite remainder and use $2\lambda = \lambda$ for nonzero limit λ. The general nonzero-limit exclusion is the published theorem [2, Theorems 1.6 and 2.6], not a new restriction specific to p-groups. Finally, an infinite p-group cannot have a d-Σ_1 Scott sentence. A true existential disjunct of its Σ_1 part has finite witnesses. They generate a finite subgroup, to which the universal part passes. This would be a nonisomorphic countable model of the same sentence. □

In the rank conventions of [4], exact d-$\Sigma_{\beta+1}$ complexity gives $\text{SR}_p = \beta < \text{SR} = \beta+1$. The displayed families therefore give explicit rank-gap examples at every countable $\beta \geq 1$. This supplies concrete examples for their Question 1; their prior abstract existence result remains credited. No uniform effective presentation from an arbitrary countable ordinal notation is being asserted.

7 Arbitrary divisible parts

The final even-form case can be treated without assuming that ACM's reduced-group statement automatically applies to a nonreduced group. We prove the sufficient fragment that is needed. Give each nonzero element of the divisible part height ∞, larger than every ordinal; as usual $h(0) = \infty$ as well. A group with infinitely many independent Prüfer summands has P_ξ infinite for every ordinal ξ.

Lemma 7.1 (Infinite-divisible-rank comparison)**.** *Let A, B be countable Abelian p-groups whose divisible ranks are both infinite. Let $g : \langle \bar{b} \rangle \to \langle \bar{a} \rangle$ be the isomorphism induced by the indicated finite tuples. The following conditions are sufficient for the displayed comparisons.*

1. *For $\delta = \omega\gamma$, $(A, \bar{a}) \leq_{2\gamma+1} (B, \bar{b})$ provided the Ulm invariants agree below δ and, for $a = g(b)$, either*

$$h_A(a) = h_B(b) < \delta, \quad \text{or} \quad h_B(b) \geq \delta, \qquad h_A(a) \geq \min\{h_B(b), \delta + \omega\}.$$

2. *$(A, \bar{a}) \leq_{2\gamma+2} (B, \bar{b})$ provided the Ulm invariants agree below $\delta = \omega\gamma$,*

$$u_B(\xi) \leq u_A(\xi) \quad (\delta \leq \xi < \delta + \omega),$$

and each $a = g(b)$ either has the same height as b below $\delta + \omega$, or both heights are at least $\delta + \omega$.

3. *At a nonzero limit η, $(A,\bar{a}) \leq_\eta (B,\bar{b})$ provided the Ulm invariants agree below $\omega\eta$ and heights are equal below $\omega\eta$ and simultaneously above that cut.*

Proof. We verify the extension argument behind the infinite-socle cases of ACM, Theorem 4.4, rather than importing its reducedness hypothesis. The following algebraic ingredients apply to arbitrary Abelian p-groups. For a finite subgroup S, write $S_\xi = S \cap G_\xi$.

If S is finite and $x \notin S$ has $px \in S$, one may choose its representative modulo S to be proper: its height is maximal in its finite coset. Among such representatives also maximize the height of its p-multiple. These two maxima exist because the coset is finite, even when the maximum is ∞. Then $h(x+s) = \min\{h(x), h(s)\}$ for every $s \in S$. If its height is ξ and $h(px) > \xi + 1$, a root z of px can be chosen in $G_{\xi+1}$, and $x - z$ is a proper socle element at height ξ. The available dimension is measured by the elementary map

$$S^*_\xi / S_{\xi+1} \longrightarrow P_\xi / P_{\xi+1}, \qquad [s] \longmapsto [s - z],$$

where $S^*_\xi = S \cap G_\xi \cap p^{-1}G_{\xi+2}$ and $z \in G_{\xi+1}$ satisfies $pz = ps$. It is injective and well-defined. A proper socle element can be chosen precisely when its image is not the entire Ulm quotient. Thus equality, or the appropriately directed inequality, of Ulm dimensions permits the usual extension at a specified ordinal height. This is the argument of ACM, Lemma 2.3; it requires no reducedness.

Finally, at a height where only a lower bound is required, infinitely many socle elements permit avoidance of the finitely many forbidden choices caused by S. Here this is available at every height from the infinite divisible part. Roots of a divisible element can be chosen divisible. More generally, to obtain a root of height at least ρ, the only assertion needed is that the given element lies in $G_{\rho+1}$. No predecessor of an infinite or limit height is taken.

Induct on the comparison index. At index zero, the given finite group isomorphism is the required atomic agreement. For an odd index $2\gamma + 1$, first fix the entire finite right extension B'. Choose a positive integer M above all k for which an element of B' has height $\delta + k$. Let r be the number of cyclic one-element extensions needed and put $N = M + r + 1$. Build B' over $\langle \bar{b} \rangle$ one element at a time, using the proper representatives with the secondary maximum just specified. After i extensions, maintain equality of heights below δ; for a right element of finite block height $\delta + k$ require its image to have height at least $\delta + k$; and for a right element of height at least $\delta + \omega$ require its image to have height at least $\delta + N - i$. The original odd condition implies this finite-headroom invariant at $i = 0$.

Below δ, a one-step height has a root of exactly the same height, and the secondary maximum prevents that root from lying in the preceding finite image. A height jump is handled by the Ulm-quotient dimension calculation above. In detail, in the one-step case put $\xi = h_B(x)$ and $y = g(px)$. A root w of y has height exactly ξ. If $w + g(s)$ had greater height for some s in the preceding subgroup, then $x + s$ would still be a proper representative of height ξ. Moreover, $p(w + g(s)) = g(p(x + s))$ would belong to $A_{\xi+2}$. Since $\xi + 2 < \delta$ and the old map preserves membership at that cut, $p(x+s) \in B_{\xi+2}$. Its height would therefore exceed $h_B(px) = \xi + 1$, contrary to the secondary maximum. Thus w is proper over the preceding image. This also rules out w in the preceding image. In the jump case choose a root w_1 of y in $A_{\xi+1}$ and a proper $w_2 \in P_\xi(A)$, using the unused Ulm-quotient dimension; $w = w_1 + w_2$ is proper of height ξ. In either case the map extends on all sums by the proper-coset identity.

For a finite block height $\delta + k$, the image of the p-multiple has height at least $\delta + k + 1$, so a root of sufficient height exists; infinitely many socle elements of at least that height permit avoidance of the finite image. For a right height at least $\delta + \omega$, use a root of height at least $\delta + N - i - 1$, again adding an unused socle element when necessary. The proper-coset identity

proves the invariant for sums whose preceding summand has a different right height; when both heights are above δ, only the indicated lower bound is needed. Thus the invariant holds on the entire new finite subgroup. At completion, all heights are equal when truncated at δ.

In particular, this step does not demand a root still above the limit $\delta + \omega$ of an element of height exactly that limit: only the explicitly finite headroom is spent.

If $\gamma = \beta + 1$, the reversed map now satisfies part (2) at index $2\beta + 2 = 2\gamma$: all invariants below δ agree and heights are equal below that cut. If γ is a limit, it satisfies part (3) at $2\gamma = \gamma$. For $\gamma = 0$, it satisfies the atomic condition. The induction hypothesis therefore gives the response required by the definition of $\leq_{2\gamma+1}$.

At an even index $2\gamma + 2$, first fix the entire finite right extension B' and put

$$M = 1 + \max\big(\{k < \omega : h_B(x) = \delta + k \text{ for some } x \in B'\} \cup \{0\}\big).$$

Let r be the number of one-element extensions needed and put $N = M + r + 1$. During the i-th extension maintain equal heights below $\delta + M$, and send every element of right height at least $\delta + \omega$ to left height at least $\delta + N - i$. The initial map satisfies this invariant. Below $\delta + M$, root and jump extensions use equality of the lower invariants and $u_B \leq u_A$ in the final block. The root/properness argument is the same as above. The Ulm-quotient calculation also applies at the greatest possible finite height $\delta + M - 1$: source elements above that height are already at least $\delta + \omega$, because there are no right-hand heights in $[\delta + M, \delta + \omega)$ anywhere in B'. Before every extension $N - i \geq M + 2$, and after all r extensions $N - r = M + 1$. Thus the old map preserves membership through $\xi + 2 \leq \delta + M + 1$, including at the boundary $\xi = \delta + M - 1$. This is the exact membership preservation used in the preceding root/properness argument as well. Hence the partial map preserves the predicates needed for S^*_ξ and $S_{\xi+1}$ at every height used in the calculation. Their dimensions agree in the two finite subgroups, and the stated Ulm inequality supplies the unused target class whenever one is needed in the source.

For a right element of height at least $\delta + \omega$, its p-multiple's image has a root of height at least $\delta + N - i - 1$; an unused socle element of at least that height avoids the preceding finite subgroup. Every right element of B' has either a height below $\delta + M$ or a height at least $\delta + \omega$, by the definition of M. For a preceding summand of height below $\delta + M$, its height and that of its image agree and are below the new root's height; the sum has that same smaller height. For a preceding summand above the cut, both image heights meet the new headroom bound, so their sum does too. This proves the invariant on all sums; nonzero coefficients modulo p do not change heights. At the end, the inverse map satisfies the odd height condition at $2\gamma + 1$. Its lower invariants agree; its left-hand socles are all infinite. The odd induction step supplies the required comparison of the extended tuples.

At a limit η, each fixed smaller successor index uses a cut strictly below $\omega\eta$. The assumed equality of invariants and truncated heights therefore supplies its sufficient conditions. Apply the induction hypothesis at a successor index above the requested smaller index to obtain the required finite response. This proves part (3) and completes the induction.

The only uses of the infinite-socle assumption in these extensions are the explicitly available avoidance choices. In particular, the reduced-case inference from length to infinite socles is unnecessary, and no exceptional finite-socle alternative is invoked. □

Theorem 7.2 (No exact pure-Σ complexity for any countable Abelian p-group)**.** *No countable Abelian p-group has exact pure Σ_α Scott complexity at any countable index.*

Proof. At index zero, every closed group-language term equals 0, so all Abelian p-groups have the same quantifier-free sentences. Consequently none has a Σ_0 Scott sentence. Write

$G = R \oplus D$. General nonzero-limit and low-index exclusions, and Theorem 4.4, leave only even-form indices $2\gamma + 2$. Put $\delta = \omega\gamma$.

If $\lambda_R \geq \delta + \omega$, the odd obstruction at $\gamma + 1$ excludes a $\Sigma_{2\gamma+3}$ Scott sentence and therefore a $\Sigma_{2\gamma+2}$ one. If $\lambda_R < \delta$, the short-length upper bound is already $\Pi_{2\gamma+1}$. If R_δ is infinite of bounded exponent, choose for each finite reduced-coordinate tuple a finite retained summand as in Lemma 4.6. Its pointed odd comparison, combined with Lemma 4.3, transfers the matrix of any putative $\Sigma_{2\gamma+2}$ Scott sentence to a nonisomorphic quotient with the same divisible part.

It remains that $\delta \leq \lambda_R < \delta + \omega$ and $F = R_\delta$ is finite, possibly zero. Suppose first that D has finite rank. The group $F \oplus D$ has a Π_2 Scott sentence. To see this, choose n killing F and impose stabilization $p^n H = p^{n+1} H$, a Π_2 sentence. For each $k \leq n$, fix the finite dimension of $(p^k H)[p]$. Having dimension at least r is Σ_1, witnessed by independent elements $p^k x_i$ with $p^{k+1} x_i = 0$; exact finite dimension is d-Σ_1. These dimensions specify all finite Ulm invariants and the finite divisible rank. Along with the Abelian p-group axioms they are countably categorical. Relativizing this Π_2 sentence to the $\Pi_{2\gamma}$ domain G_δ gives a $\Pi_{2\gamma+2}$ Scott sentence after the lower Ulm list is included. If a $\Sigma_{2\gamma+2}$ Scott sentence exists too, Miller's theorem lowers it to d-$\Sigma_{2\gamma+1}$. Thus pure $\Sigma_{2\gamma+2}$ is not exact.

Suppose instead that D has infinite rank. Replace the finite terminal Ulm factor F by $F \oplus C_p$, leaving every factor before γ unchanged. If $F = 0$, append the new final factor C_p. The earlier factors are unbounded because $\lambda_R \geq \delta$. For $\gamma > 0$, Zippin's theorem supplies a reduced group R' with this sequence; for $\gamma = 0$ take $R' = F \oplus C_p$ directly. Put $G' = R' \oplus D$. The Ulm invariants of G' and G agree below δ; in the block from δ to $\delta + \omega$, the former dominate the latter, with strict inequality at δ. Both groups have infinite divisible rank. Lemma 7.1(2), with left group G' and right group G and empty tuples, gives

$$G' \leq_{2\gamma+2} G.$$

Every $\Sigma_{2\gamma+2}$ sentence true of G is therefore true of G', whereas their Ulm invariants differ. Hence G has no $\Sigma_{2\gamma+2}$ Scott sentence in this last case. All cases are exhausted. □

8 Answers to the source questions and remaining work

Theorem 2.1 gives a negative answer to [1, Question 5.1]. Infinite Ulm invariants below the terminal block do not force the generic upper bound to be optimal. For a concrete example use

$$G = B_\omega \oplus W_\omega^{(\omega)}.$$

Its finite-height Ulm invariants and its ωth invariant are all ω, and its length is $\omega + 1$. Nevertheless its terminal group is $C_p^{(\omega)}$, so its exact complexity is Π_4, whereas the generic case-(ii) bound of [1, Theorem 3.1] is Π_5. More precisely, that generic bound is optimal exactly when at least two terminal cyclic multiplicities are infinite. At exact limit length the $\Pi_{2\gamma+1}$ bound is always optimal; for nonzero finite tails, the generic difference bound improves by one index.

Theorem 7.2 and Theorem 6.2 answer both realization alternatives of [1, Question 5.2]: the pure-Σ spectrum is empty, and the difference spectrum consists of all positive countable successor levels. The impossibility of exact limit difference levels is the general published restriction, while the successor realizations already lie in the reduced subclass.

The optimal complexity of each individual nonreduced group is not classified here. Nor do the special comparison lemmas eliminate all relative-sentence conditions from the general

pointed back-and-forth analysis requested in [1, Question 5.3]. First-order Vaught's conjecture is not resolved by these Scott-sentence results.

Preparation and status. This manuscript was prepared with AI assistance from a research programme directed by the repository owner. The assembly combines the argument for successor realizations, its transfinite-existence repair, the divisible-rank argument, and the terminal classification into one canonical text. Published dependencies are credited at use. The unbounded assertions rest on the displayed hand proofs and those published inputs; no independent human audit, formal verification, or historical priority claim is asserted. No private notes, named project claims, or conventions outside this repository are required.

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